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Project Report

Design and Implementation of a VLAN-Based Branch Network Using Cisco Packet Tracer

Course: Computer Networks Lab
Course Code: ICT-3104

Submitted To

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1. Objective

XYZ Company, headquartered in Dhaka, is establishing a new branch near Savar and requires a small, self-contained Local Area Network (LAN) for the new office. This report presents the design and implementation of the network, including the IP addressing scheme, VLAN segmentation, wireless connectivity, DHCP configuration, and network verification. The entire network is designed using a single Cisco router and a single Cisco switch and is implemented and tested in Cisco Packet Tracer to ensure it meets the branch's operational requirements.

2. Requirements

The branch network was designed to satisfy the following requirements set by the company:

- Use exactly one Cisco router and one Cisco switch per branch.
- Support three departments: Admin/IT, Finance/HR, and Customer Service/Reception.
- Place each department on its own VLAN, logically separating broadcast domains.
- Provide wireless connectivity for users in every department.
- Assign IPv4 addresses to all host devices automatically via DHCP.
- Allow devices in all three departments to communicate with one another (inter-VLAN routing).
- Operate independently of the Dhaka headquarters network, using the ISP-assigned base network 192.168.1.0.

3. IP Addressing Scheme

The ISP-assigned base network 192.168.1.0/24 was subnetted using variable-length subnet masking (VLSM) into three equal subnets of 64 addresses each (/26, subnet mask 255.255.255.192), one per department. This provides ample headroom for PCs, laptops, smartphones, printers, and the access point in each department, while keeping the three departments in separate broadcast domains.

Department	VLAN	Network Address	Subnet Mask	Usable Host Range	Default Gateway
Admin/IT	10	192.168.1.0/26	255.255.255.192	192.168.1.1 – .62	192.168.1.1
Finance/HR	20	192.168.1.64/26	255.255.255.192	192.168.1.65 – .126	192.168.1.65
CS/Reception	30	192.168.1.128/26	255.255.255.192	192.168.1.129 – .190	192.168.1.129

4. Network Topology

A router-on-a-stick design was used: a single physical link from the router's GigabitEthernet0/0 interface to the switch's Fa0/1 port carries all three VLANs as an 802.1Q trunk. The switch provides access ports for wired hosts and for a wireless access point per department, and the router's subinterfaces perform inter-VLAN routing.

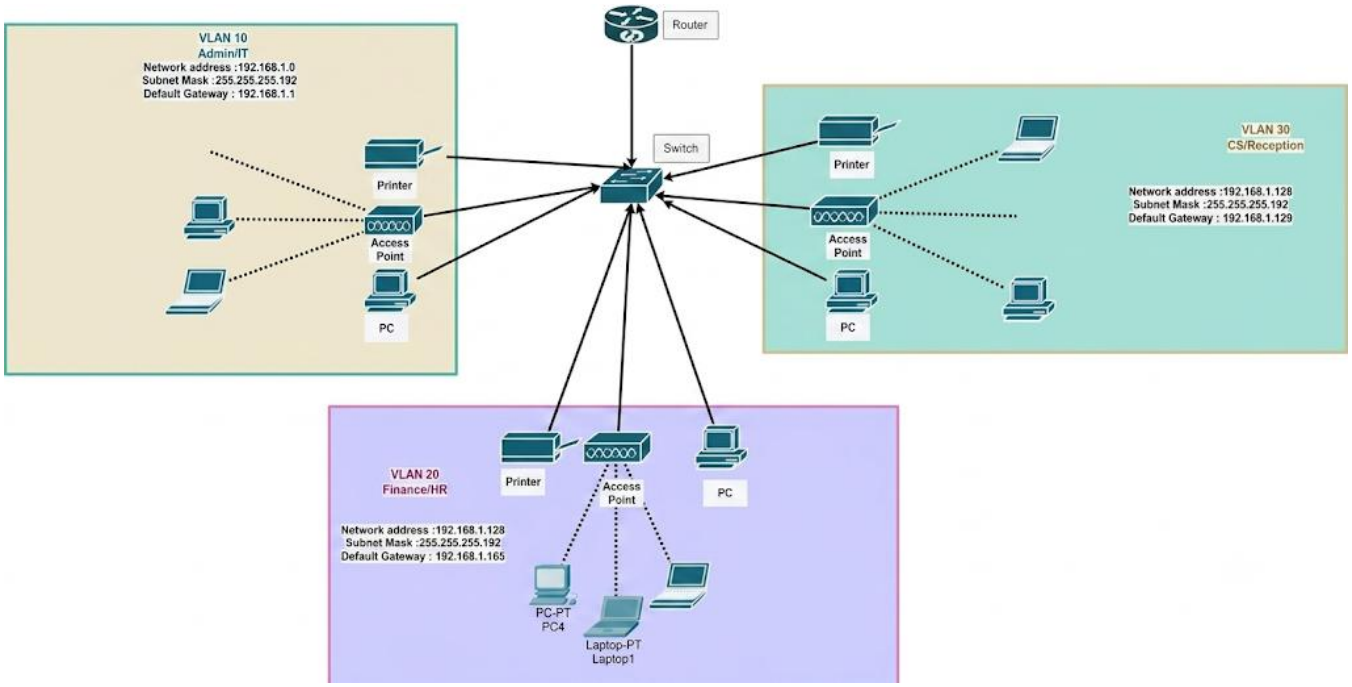


Figure 1: Logical network diagram showing VLAN 10 (Admin/IT), VLAN 20 (Finance/HR), and VLAN 30 (CS/Reception), each with a PC, laptop, smartphone, printer, and access point connected through the switch to the router.

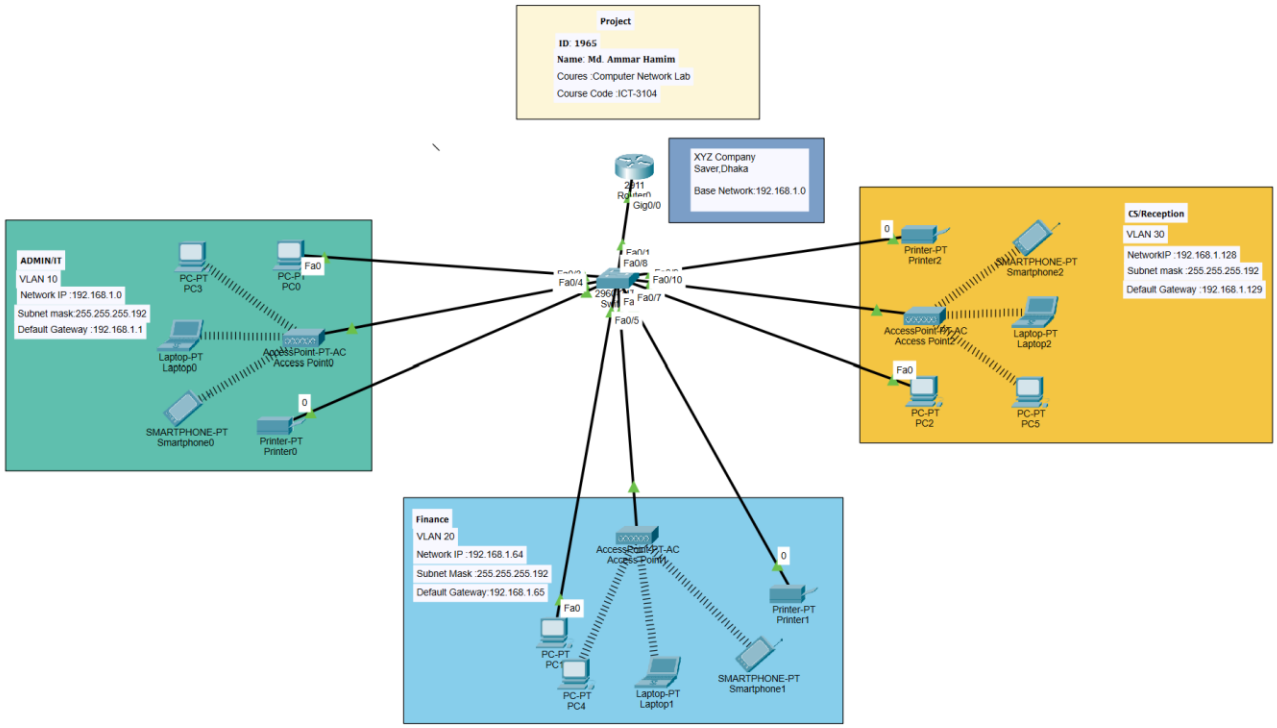


Figure 2: Implemented topology in Cisco Packet Tracer, showing the router, switch, VLAN colour-coded departments, and per-VLAN addressing details.

5. VLAN Design and Switch Configuration

Three VLANs were created on the switch, one per department, and access ports were assigned accordingly. Fa0/1 was configured as an 802.1Q trunk to carry all VLAN traffic to the router.

VLAN ID	VLAN Name	Department	Assigned Switch Ports
10	VLAN0010	Admin/IT	Fa0/2 – Fa0/4
20	VLAN0020	Finance/HR	Fa0/5 – Fa0/7
30	VLAN0030	CS/Reception	Fa0/8 – Fa0/10

5.1 Switch Configuration Commands

```
Switch#configure terminal

! Create VLANs and assign access ports
Switch(config)#interface range fa0/2-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10

Switch(config)#interface range fa0/5-7
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20

Switch(config)#interface range fa0/8-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30

! Configure the uplink to the router as an 802.1Q trunk
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#write memory
```

Verification with show vlan brief confirms the three VLANs are active with the correct ports assigned:

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   VLAN0010                         active    Fa0/2, Fa0/3, Fa0/4
20   VLAN0020                         active    Fa0/5, Fa0/6, Fa0/7
30   VLAN0030                         active    Fa0/8, Fa0/9, Fa0/10
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 3: Output of show vlan brief confirming VLAN 10, 20, and 30 with their respective access ports.

6. Router Configuration (Router-on-a-Stick)

The router's physical interface GigabitEthernet0/0 was enabled and divided into three logical subinterfaces, one per VLAN. Each subinterface was configured with 802.1Q encapsulation matching the VLAN ID and given the default-gateway address of its department's subnet, enabling inter-VLAN routing across all three departments.

```
Router#configure terminal

Router(config)#interface gigabitEthernet 0/0
```

```
Router(config-if)#no shutdown
Router(config-if)#exit

Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.1.1 255.255.255.192
Router(config-subif)#exit

Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.1.65 255.255.255.192
Router(config-subif)#exit

Router(config)#interface gigabitEthernet 0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.1.129 255.255.255.192
Router(config-subif)#end
Router#write memory
```

The show ip interface brief output confirms all three subinterfaces are up/up with the correct IP addresses, providing the gateway for each VLAN:

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.1.1     YES manual up                    up
GigabitEthernet0/0.20  192.168.1.65    YES manual up                    up
GigabitEthernet0/0.30  192.168.1.129   YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

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Figure 4: show ip interface brief showing GigabitEthernet0/0.10, .20 and .30 up/up with their configured gateway addresses.

7. DHCP Configuration

The router was configured as the DHCP server for all three departments so that every host device — PCs, laptops, smartphones, and access-point clients — obtains its IPv4 configuration automatically, satisfying the requirement for automatic addressing.

```
Router(config)#service dhcp

Router(config)#ip dhcp pool Admin-Pool
Router(dhcp-config)#network 192.168.1.0 255.255.255.192
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 192.168.1.1
Router(dhcp-config)#domain-name Admin.com
Router(dhcp-config)#exit

Router(config)#ip dhcp pool Finance-Pool
Router(dhcp-config)#network 192.168.1.64 255.255.255.192
Router(dhcp-config)#default-router 192.168.1.65
Router(dhcp-config)#dns-server 192.168.1.65
Router(dhcp-config)#domain-name Finance.com
Router(dhcp-config)#exit

Router(config)#ip dhcp pool CS-Pool
Router(dhcp-config)#network 192.168.1.128 255.255.255.192
Router(dhcp-config)#default-router 192.168.1.129
Router(dhcp-config)#dns-server 192.168.1.129
Router(dhcp-config)#domain-name CS.com
```

```
Router(dhcp-config)#end
```

On a wireless client, the DHCP request succeeded, automatically receiving an address, subnet mask, and gateway from the Admin/IT pool:

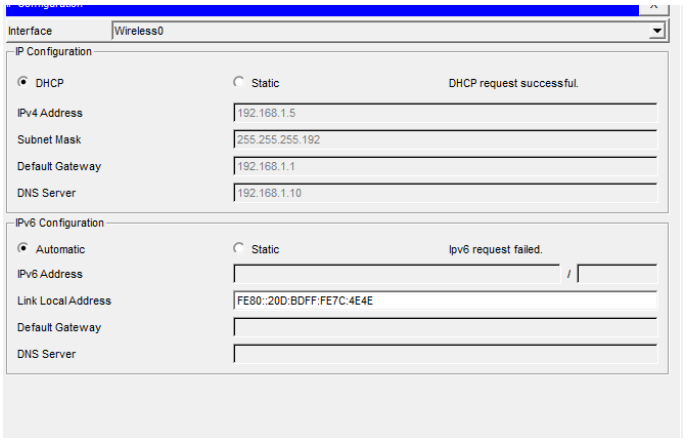


Figure 5: A wireless host successfully receiving 192.168.1.5/26 via DHCP with gateway 192.168.1.1.

Reviewing the router's lease table confirms multiple automatic bindings issued across all three subnets:

```
Router>
Router>en
Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.1.2      000A.F39D.D181           --                     Automatic
192.168.1.3      0005.5E64.3669           --                     Automatic
192.168.1.6      00D0.D392.93A8           --                     Automatic
192.168.1.5      000D.BD7C.4E4E           --                     Automatic
192.168.1.4      0001.97B3.E061           --                     Automatic
192.168.1.66     00D0.9701.7745           --                     Automatic
192.168.1.67     0030.A3B1.9E46           --                     Automatic
192.168.1.69     0030.F282.7D88           --                     Automatic
192.168.1.70     000C.CF44.DB32           --                     Automatic
192.168.1.131    0060.2F77.7185           --                     Automatic
192.168.1.132    0040.0B29.5045           --                     Automatic
192.168.1.130    0002.176E.B35B           --                     Automatic
192.168.1.134    00D0.5809.0A1B           --                     Automatic
192.168.1.135    0050.0F77.81C2           --                     Automatic
Router#
```

Figure 6: show ip dhcp binding listing automatic leases issued to hosts in all three departments.

8. Wireless Access Configuration

Each department was equipped with a Cisco access point to satisfy the requirement that every department has wireless connectivity. Every AP was assigned a department-specific SSID (Admin-WiFi, Finance-WiFi, and CS-WiFi) and secured with WPA2-PSK/AES authentication, and end devices were fitted with wireless network-interface modules (Linksys-WMP300N for desktops, Linksys-WPC300N for laptops) to connect to these networks.

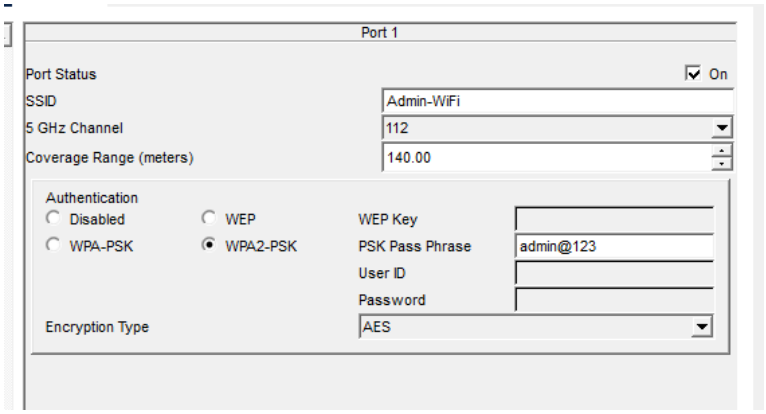


Figure 7: Access point Port 1 configuration — SSID "Admin-WiFi" secured with WPA2-PSK and AES encryption.

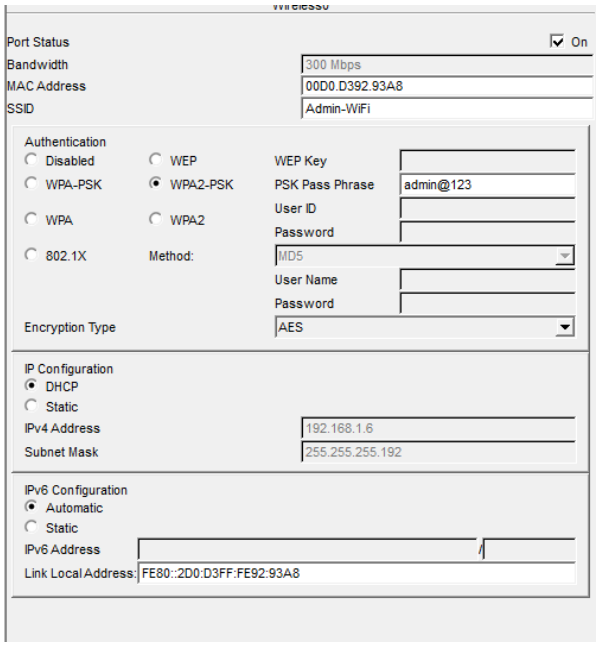


Figure 8: Wireless client adapter configuration matching the SSID and WPA2-PSK passphrase, obtaining 192.168.1.6/26 via DHCP.

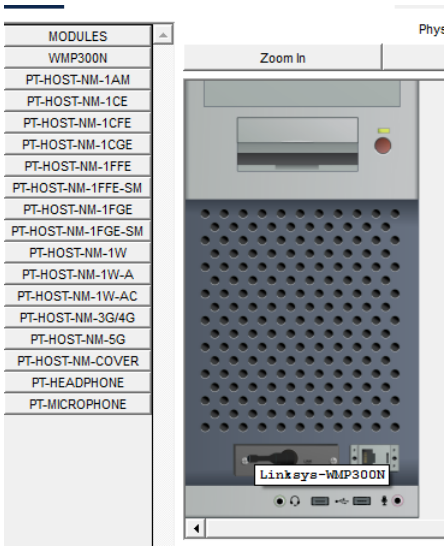


Figure 9: PT-HOST-NM-1W wireless module (Linksys-WMP300N) installed on a desktop PC to enable wireless connectivity.

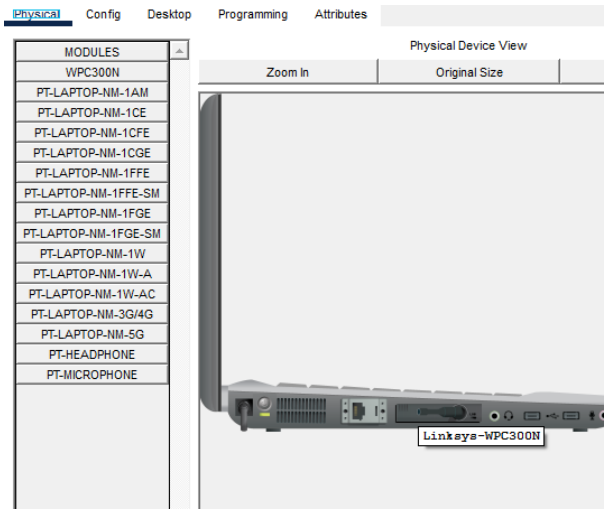


Figure 10: Wireless module (Linksys-WPC300N) installed on a laptop for department wireless access.

From an end-user device, all three department SSIDs are visible and reachable, each secured with WPA2-PSK:

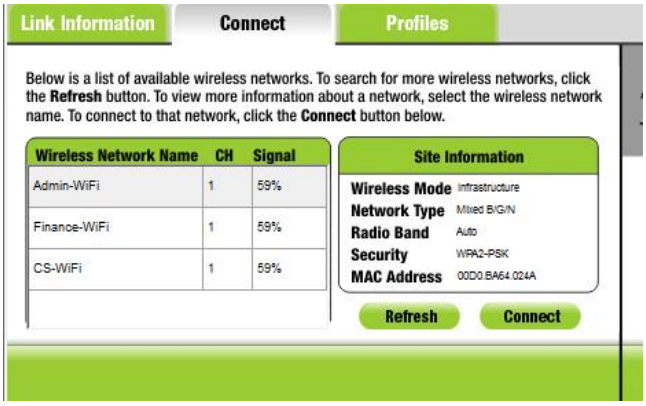


Figure 11: Wireless client scan showing Admin-WiFi, Finance-WiFi, and CS-WiFi, all using WPA2-PSK security.

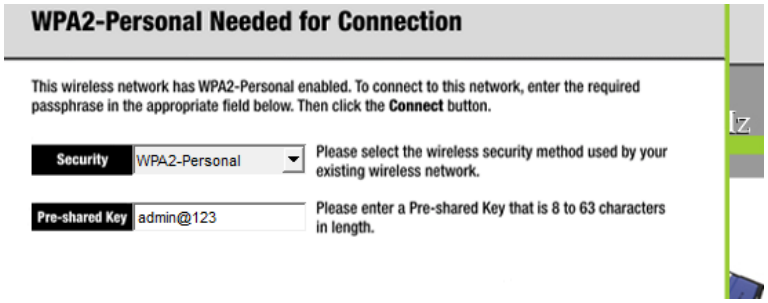


Figure 12: Connection prompt requesting the WPA2-Personal pre-shared key before joining the department wireless network.

9. Testing and Verification

Connectivity was verified both within a VLAN and across VLANs to confirm the network meets the requirement that devices in all departments can communicate with each other.

9.1 Ping Test — Admin/IT to Finance/HR (Inter-VLAN)

An ICMP ping from PC3 (VLAN 10, Admin/IT) to Smartphone1 (VLAN 20, Finance/HR) succeeded, confirming that the router correctly routes traffic between VLANs:

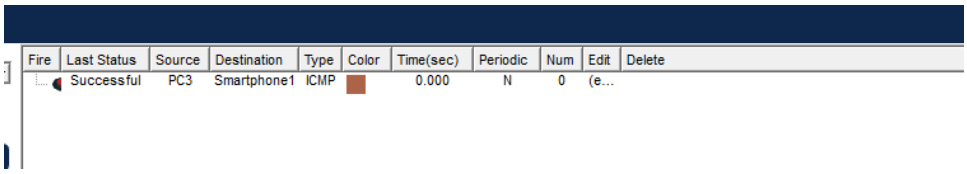


Figure 13: Simulation event list showing a successful ICMP exchange between PC3 (Admin/IT) and Smartphone1 (Finance/HR).

9.2 Sustained Ping Test Across the Network

A sustained ping to a wireless host address (192.168.1.134) returned four successful replies with 0% packet loss, confirming stable connectivity for wireless devices reached over the routed network:

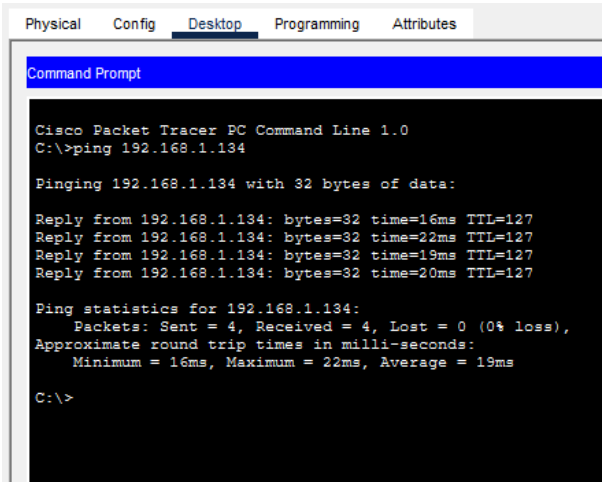


Figure 14: Command-line ping to 192.168.1.134 — 4 packets sent, 4 received, 0% loss.

Together, these tests confirm that hosts in Admin/IT, Finance/HR, and CS/Reception — whether wired or wireless — can reach one another through the router's inter-VLAN routing, and that DHCP-assigned addresses are fully functional.

10. Requirements Traceability

Requirement	Status	Evidence
One router and one switch per branch	Met	Single router (Gig0/0 trunk) and single switch topology (Fig. 1, 2)
Three departments: Admin/IT, Finance/HR, CS/Reception	Met	VLAN 10, 20, 30 configuration (Section 5)
Each department in a different VLAN	Met	show vlan brief output (Fig. 3)
Wireless network per department	Met	Per-department APs with unique SSIDs (Section 8)
Automatic IPv4 addressing	Met	DHCP pools and lease bindings (Section 7)
Inter-department communication	Met	Successful inter-VLAN ping tests (Section 9)

11. Conclusion

The branch network for XYZ Company's Savar location was successfully designed and implemented in Cisco Packet Tracer using a single Cisco router and a single Cisco switch. The ISP-assigned **192.168.1.0/24** network was subnetted into three **/26** subnets, with one subnet allocated to each department. Each department was placed in a separate VLAN to enhance security and reduce broadcast traffic. Inter-VLAN communication was enabled through router-on-a-stick configuration using router subinterfaces, while DHCP pools automatically assigned IP addresses to all host devices. Dedicated wireless access points provided secure wireless connectivity for each department. Network verification using **show** commands and ping tests confirmed that all project requirements were successfully met, and the branch network operates independently of the Dhaka headquarters network as specified.